ME 333

MECHANICS OF MATERIALS

Lecture 20: Unsymmetric Bending III & Shear Center

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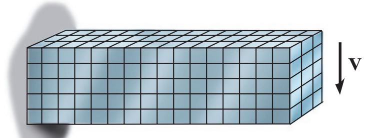
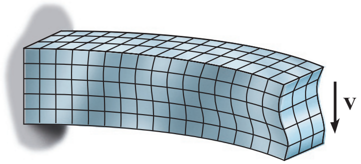
Stresses due to varying bending moments

Cantilever beam with shear force acting on the free end



(a) Before deformation



(b) After deformation

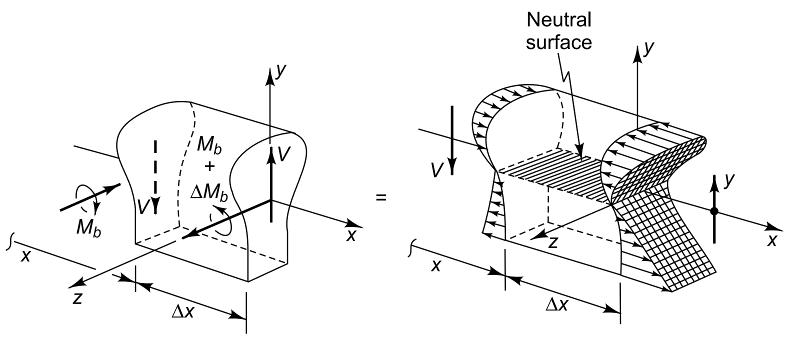
Assumptions :

- Bending stress distribution is valid even when the bending moment varies along the beam in presence of shear forces
- Satisfaction of geometric compatibility is omitted in the analysis - engineering theory
- Satisfaction of stress-strain relations is not included

Ref Base Fig: Statics & Mechanics of Materials, R. C. Hibbeler

Stresses due to varying bending moments

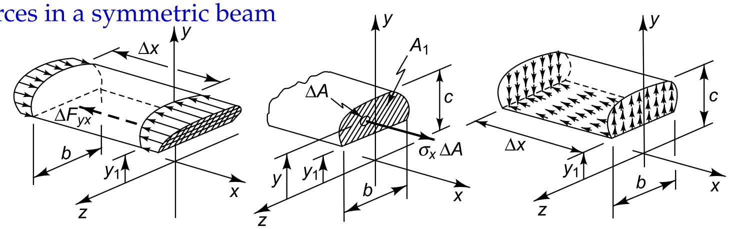
Shear forces in a symmetric beam



Ref Base Figs: An Introduction to Mechanics of Solids, Crandall et. al.

Stresses due to varying bending moments

Shear forces in a symmetric beam



- Using static force equilibrium:

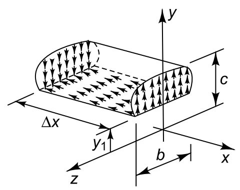
$$\Delta F_{yx} = -\frac{\Delta M_b}{I_{zz}} \int_A y\, dA$$

$$\frac{dF_{yx}}{dx} = \lim_{\Delta x \to 0} \frac{\Delta F_{yx}}{\Delta x} = -\frac{dM_b}{dx} \frac{1}{I_{zz}} \int_A y\, dA = \frac{V}{I_{zz}} \int_A y\, dA$$

Ref Base Figs: An Introduction to Mechanics of Solids, Crandall et. al.

Shear stress distribution

Shear Flow q_{yx} is the resultant of shear stress τ_{yx} distributed across the width b of the beam.



Shear stress is max at the neutral surface and falls off parabolically.

- Assuming shear stress to be uniform across the beam :

$$\tau_{yx} = \frac{q_{yx}}{b} = \frac{VQ}{bI_{zz}} @ y = y_1$$

- Using Equilibrium equation:

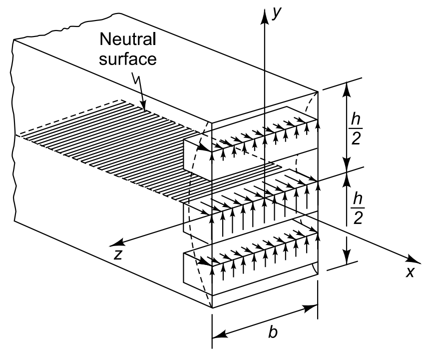
$$\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} = 0 \implies -\frac{\partial \tau_{xy}}{\partial y} = \frac{\partial}{\partial x}\left(\frac{-M_b\, y}{I_{zz}}\right) = \frac{V}{I_{zz}} y$$

$$\therefore -\int_{y_1}^{h/2} \frac{\partial \tau_{xy}}{\partial y}\, dy = \frac{V}{I_{zz}} \int_{y_1}^{h/2} y\, dy \implies -[\tau_{xy}]_{y_1}^{h/2} = \frac{V}{I_{zz}} [y^2/2]_{y_1}^{h/2}$$

$$\therefore \tau_{xy} = \frac{V}{2I_{zz}} \left[(h/2)^2 - y^2\right]$$

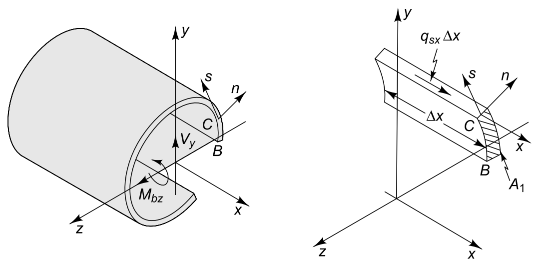
Shear stress distribution

Rectangular cross-sections : Shear stress is max at the neutral surface and falls off parabolically.



Unsymmetric Bending

In presence of shear forces (Open thin wall sections)



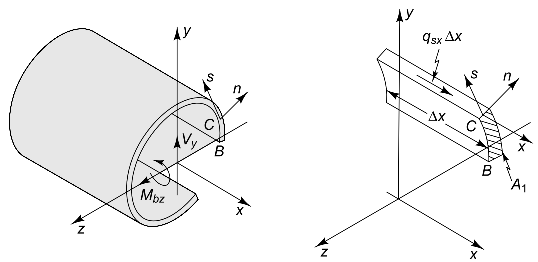
Assumption :

The stress distribution due to bending moments still remains:

$$\sigma_x = -\frac{(yI_{yy} - zI_{yz})M_z + (yI_{yz} - zI_{zz})M_y}{(I_{zz}I_{yy} - I_{yz}^2)}$$

Unsymmetric Bending

In presence of shear forces (Open thin wall sections)

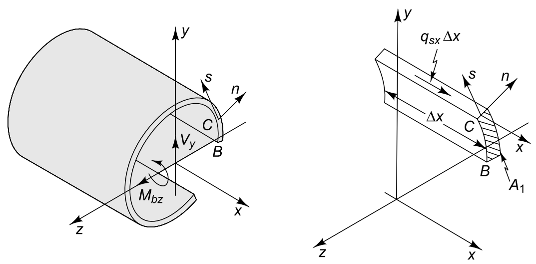


Applying static force equilibrium on segment BC :

$$q_{xs} = q_{sx} = \frac{-V_y}{I_{zz}I_{yy} - I_{yz}^2}\left(I_{yy}\int_{A_1} y\, dA - I_{yz}\int_{A_1} z\, dA\right)$$

Unsymmetric Bending

In presence of shear forces (Open thin wall sections)



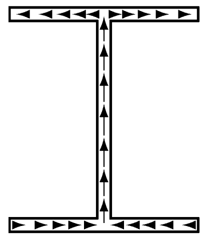
$$q_{xs} = q_{sx} = \frac{-V_y}{I_{zz}I_{yy} - I_{yz}^2}\left(I_{yy}\int_{A_1} y\,dA - I_{yz}\int_{A_1} z\,dA\right)$$

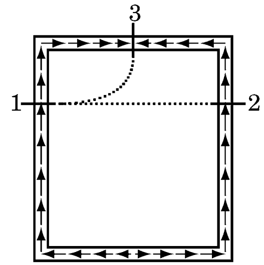
$\boxed{Q}$ Where should the shear force act ?

Ref Base Figs: An Introduction to Mechanics of Solids, Crandall et. al.

Unsymmetric Bending

$\boxed{Q}$ Why open thin sections only ?

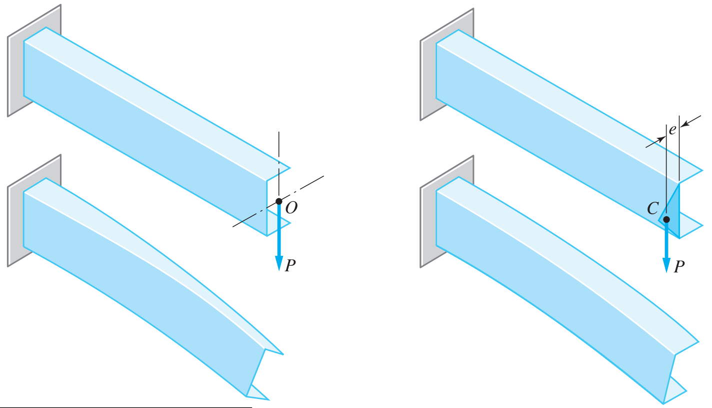




Ref Base Fig: Introduction to Solid Mechanics, Lubliner & Papadopoulos

Shear Center

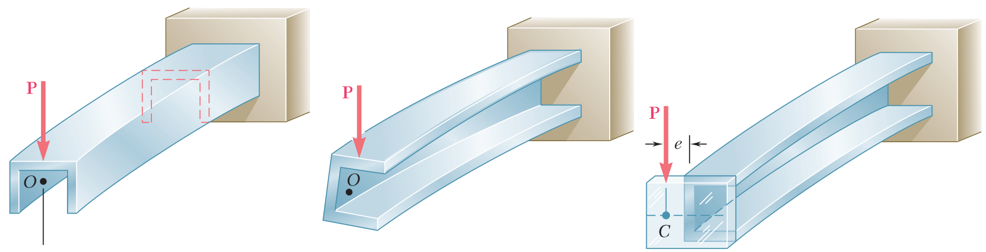
Loading of open thin sections



Ref Base Fig: Mechanics of Materials, Pytel & Kiusalaas

Shear Center

Loading of open thin sections

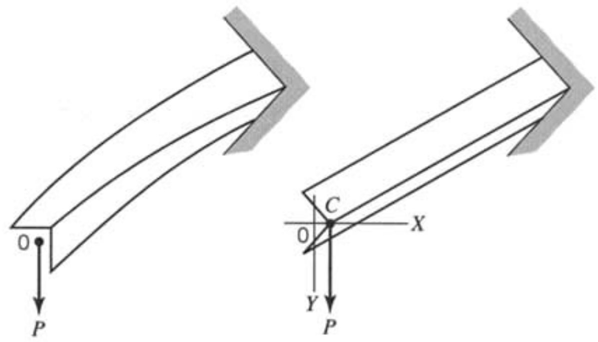


Shear Center (C) : It is the point about which if an external load is applied - it produces no twisting moment.

Ref Base Fig: Mechanics of Materials, Beer & Johnston

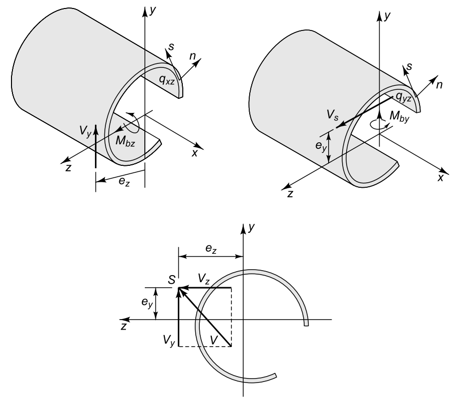
Unsymmetric Bending

Loading of open thin sections



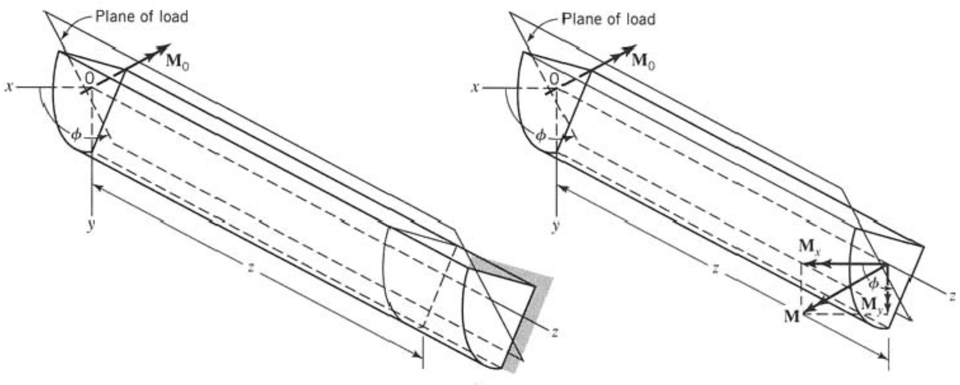
Unsymmetric Bending

Loading of open thin sections



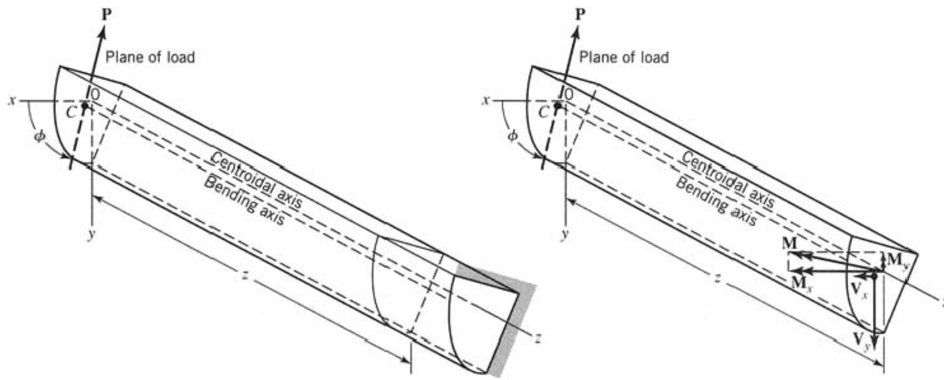
Unsymmetric Bending

Pure Bending Scenario:



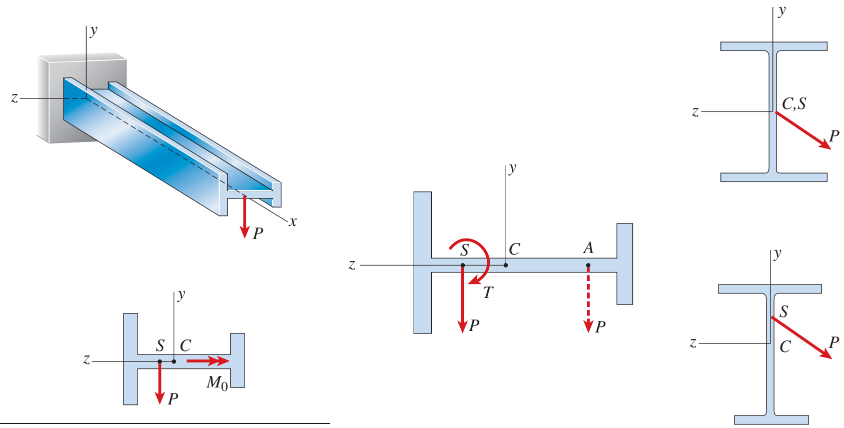
Unsymmetric Bending

Varying Bending Scenario:



Shear Center

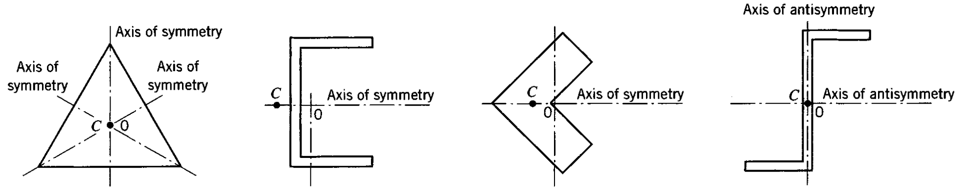
Centroid & Shear Center:



Ref Base Fig: Mechanics of Materials, Gere & Goodno

Unsymmetric Bending

Beam cross-sections:



- When there are atleast two axis of symmetry or anti-symmetry the centroid of the cross-section coincides with the shear center.
- If there is an axis of symmetry the shear center lies on the axis of symmetry.

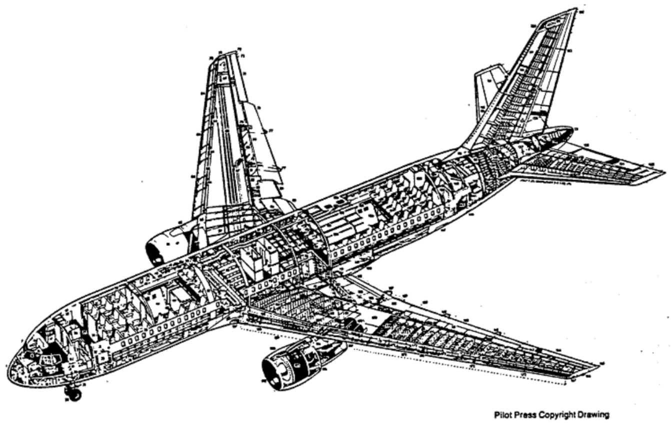
$\boxed{Q}$ How to locate the shear center ?

Ref Base Fig: Advanced Mechanics of Materials, Boresi & Schmidt

Shear Center : Applications

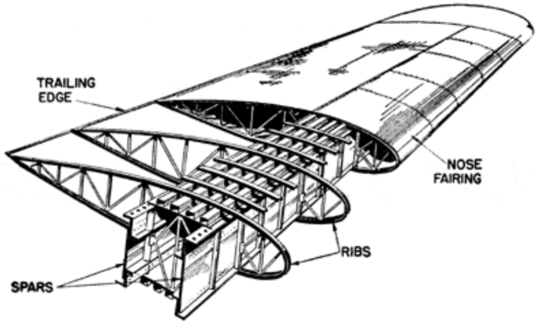
Ref Fig: Web

Shear Center : Applications



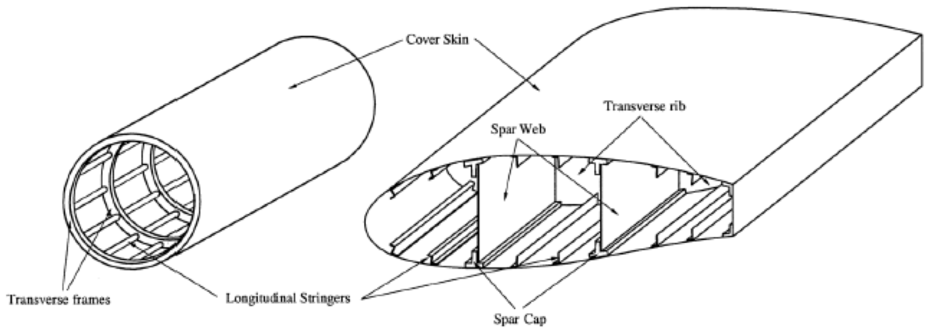
Ref Fig: Web

Shear Center : Applications



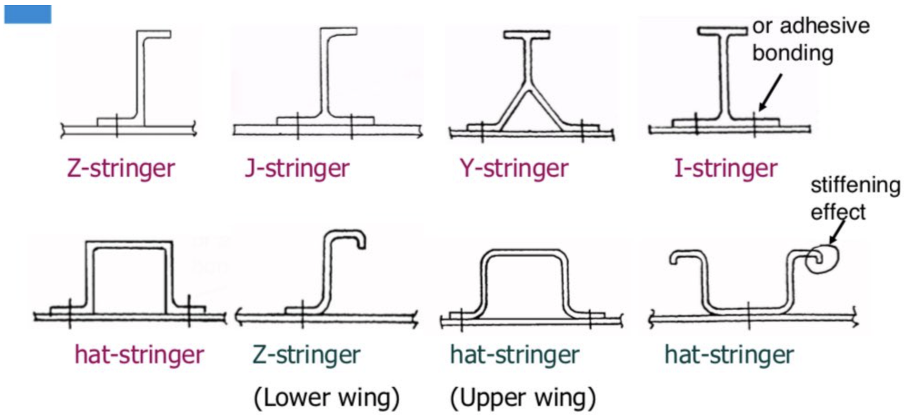
Ref Fig: Web

Shear Center : Applications



Ref Fig: Web

Shear Center : Applications



Ref Fig: Web

Reading Assignment

Chapter 7 : Crandall et. al.
Chapters 7 & 8 : Boresi & Schmidt